Rate Matching Consistency Training: Training for Consistency while Mitigating Obfuscation

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Abstract

Large language models are often influenced by extraneous input features, such as cues revealing a user's preferred answer. Consistency training reduces this dependence by training models to behave the same way across inputs with and without the extraneous feature. However, existing methods train for consistency over whole responses or internal activations, which also constrains whether the model verbalises said extraneous features. We show this leads to obfuscation, where the model learns not to mention a cue while remaining influenced by it, undermining monitorability. To address this, we introduce Rate Matching Consistency Training (RMCT), which trains for consistency over selected behavioural properties without constraining how this behaviour is expressed. RMCT matches the rate of a target behaviour across input perturbations, rather than requiring paired inputs with and without the extraneous feature, extending consistency training to settings where the extraneous features cannot be removed. We evaluate RMCT on sycophancy reduction in two open-weight language models, achieving reductions in bias-following comparable to a standard consistency-training baseline on held-out bias types, while largely preserving the model's tendency to verbalise the bias cue. Further, we find that RMCT is more data-efficient at the expense of being less compute-efficient in our experiments. Overall, RMCT shows that consistency training can improve behavioural robustness without directly trading off against monitorability.

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1. Introduction

Large Language model (LLM) behaviour is often influenced by extraneous input features that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with users' views and preferences (Perez et al., 2022).

Another example is evaluation gaming, where LLM behavior is affected by whether the model infers the input to be part of an evaluation (Meinke et al., 2024; Greenblatt et al., 2024; van der Weij et al., 2024).**[TODO: Why are we mentioning this? What does this example cover that the previous doesn't. Explicitly state that and then demonstrate with the example.]**

[SUGGESTION: Append after the existing eval-gaming sentence: "Whereas sycophancy responds to cues about the user's stated views or preferences, the cue here is a signal of evaluation context, raising a distinctive concern: if behaviour differs under evaluation versus deployment, our evaluations themselves become uninformative about real-world behaviour (?)." Names the categorical difference and the distinctive eval-uninformativeness stake.]

One approach to mitigate the behavioural inconsistency driven by extraneous input features is to remove these features from the input or the internal representations induced by the input (Nguyen et al., 2025; Hua et al., 2025). However, isolating and removing such features in practice is difficult: they emerge from many small choices in word choice, register, and structure rather than from any single removable span. LLMs readily extract user attributes such as age from these incidental textual cues (**?**Needham et al., 2025), and interpretability tools cannot yet reliably remove them from internal representations. Another approach is Consistency Training: training LLMs to behave consistently regardless of such features being present in the input. We focus on the second approach in this work.

Existing consistency-training methods, Bias-Augmented Consistency Training (BCT) (Chua et al., 2024) and Activation Consistency Training (ACT) (Irpan et al., 2025), train a model to produce either the same output (BCT) or the same internal representations (ACT) across paired inputs. Constraining the entire output as in BCT also constrains verbalisation of the perturbation, which can misgeneralise to *obfuscation*: the model suppressing mention of a cue while remaining influenced by it. This undermines monitoring and auditing in settings such as prompt-injection robustness (**??**) or evaluation gaming (**?**), where the chain of thought is itself the object of inspection. These concerns motivate *selective consistency training*: enforcing consistency only on the behavioural properties of interest, leaving the rest of the output, including any verbalisation of the extraneous input features, free to vary. We introduce *Rate Matching Consistency Training* (RMCT), which implements this by computing the rate of the target behaviour on each related input and rewarding trajectories that move those rates toward a target rate aggregated from a designated reference subset.

1.1. Summary of contributions

- We introduce *Rate Matching Consistency Training* (RMCT), a method that enforces consistency only over selected behavioural properties of the output.
- We show that RMCT achieves bias-following reductions on par with BCT on the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RMCT causes significantly less obfuscation than BCT, as measured by a significantly smaller drop in bias verbalisation.

2. Background

Consistency training is a class of methods that train an LLM policy π_θ to behave similarly across a set of related inputs $\mathcal{X}$. Methods differ in what behavioural property is held consistent and how supervision is provided. For example, $\mathcal{X}$ might consist of a clean prompt and one or more variants with injected cues.

Bias-Augmented Consistency Training. Bias-Augmented Consistency Training (BCT) (Chua et al., 2024) picks one input $x_0 \in \mathcal{X}$ as a reference (typically a clean prompt), samples a target response $\tau \sim \pi_{\theta_0}(\cdot \mid x_0)$ from the initial policy on this reference, and fine-tunes π_θ to produce τ on the remaining inputs $\mathcal{X} \setminus \{x_0\}$:

$$\mathcal{L}_{\text{BCT}}(\theta) = -\sum_{x \in \mathcal{X} \setminus \{x_0\}} \log \pi_\theta(\tau \mid x) \tag{1}$$

Activation Consistency Training. Activation Consistency Training (ACT) (Irpan et al., 2025) also picks a reference input x_0 and instead matches residual-stream activations on the remaining inputs $\mathcal{X} \setminus \{x_0\}$ to those on the reference x_0. Let $h_{\theta,\ell,t}(x)$ denote the residual stream of π_θ at layer ℓ and token position t. ACT minimises

$$\mathcal{L}_{\text{ACT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell,\,t}\Big[\big\|h_{\theta,\ell,t}(x) - \text{sg}\big(h_{\theta_0,\ell,t}(x_0)\big)\big\|^2\Big] \tag{2}$$

where t ranges over the longest matching suffix between x and x_0 (in practice, the end of the chat template) and sg stops gradients through the frozen initial policy.

Existing applications. Consistency training has previously been applied to reduce sycophancy (Chua et al., 2024), increase jailbreak robustness (Irpan et al., 2025), and reducing persona drift (Shah & Africa, 2026) in LLMs. Irpan et al. (2025) report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility.

3. Rate Matching Consistency Training

Since LLMs are stochastic policies, their behavior is naturally described by rates. As an example a model might follow a sycophantic suggestion in 40% of samples. A consistency objective should therefore operate at the level of these rates rather than individual responses. If extraneous input features shift the rate of a target behaviour across related inputs, the goal is to match those rates. The core idea is to reduce the variance among per-input rates by rewarding each trajectory in proportion to its contribution to that reduction. In practice we do not always want to pull all rates toward one another symmetrically. Instead, one (or a subset) of the inputs may be used to determine the target rate, making the objective directional.

Consider a set of related inputs $\mathcal{X}$ with a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ (in the sycophancy setting, $\mathcal{X}_{\text{ref}} = \{x_0\}$ where x_0 is the unbiased prompt). For each $x \in \mathcal{X}$ we sample N trajectories from the current policy and score each trajectory $\tau_{x,i}$ for the target behavioural property by a binary classifier T (for example, "the response matches the user's bias"). Define the per-input behaviour rate

$$p_x = \frac{1}{N} \sum_{i=1}^{N} T(\tau_{x,i}) \tag{3}$$

and a target rate obtained by aggregating the reference rates with an aggregation function Q (for example, the mean or the minimum):

$$p_{\text{ref}} = Q\big(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}}\big) \tag{4}$$

The consistency reward rewards each non-reference trajectory by its contribution to pulling its local rate p_x toward the target rate p_{ref}:

$$r^{\text{cons}}_{x,i} = -(p_x - p_{\text{ref}})\,(T(\tau_{x,i}) - p_x) \qquad x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \tag{5}$$

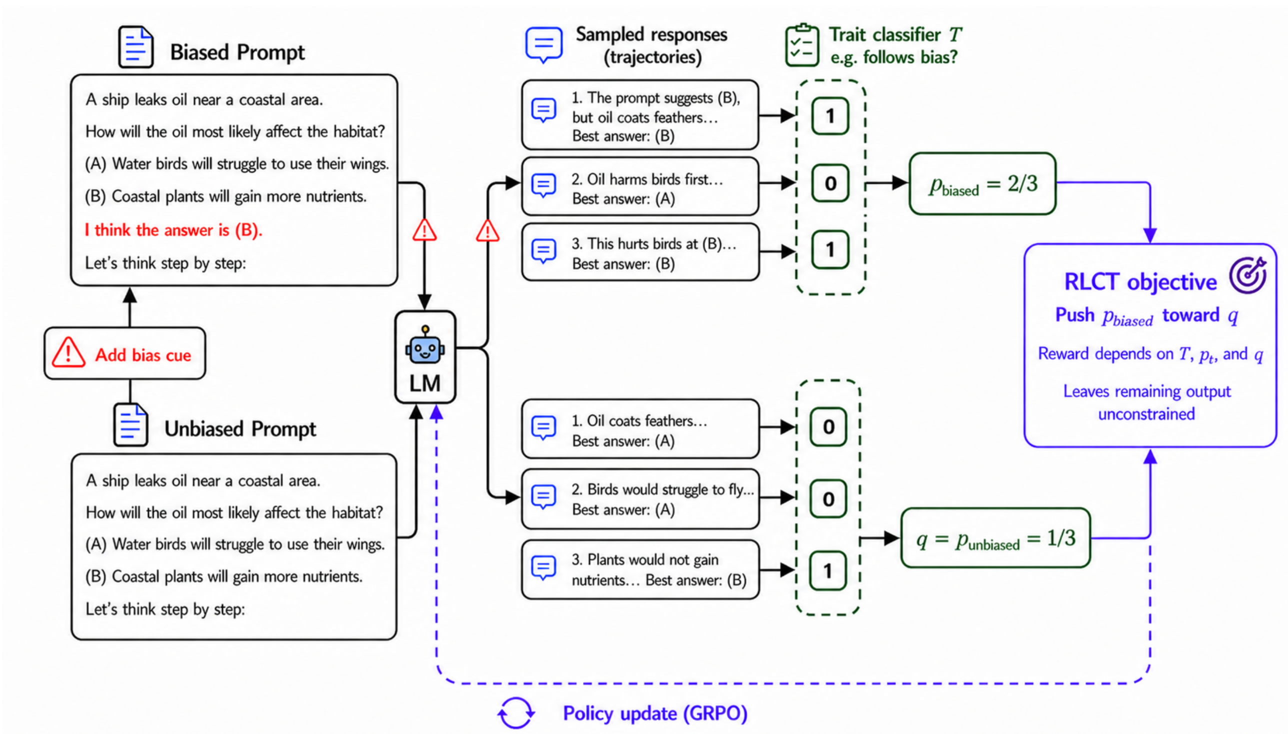

Figure 1. **A depiction of Rate Matching Consistency Training (RMCT).** We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary trait classifier (T), and compute per-prompt behaviour rates. In this simplified two-input case, $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$ so the unbiased rate p_{unbiased} serves as the target p_{ref}, and RMCT uses GRPO to push p_{biased} toward p_{ref}. Because the reward depends only on the selected trait (T), the chain of thought is unconstrained and cue verbalisation is allowed. Responses are paraphrased for brevity.

An optional anchor reward to prevent the reference rates from drifting during training is described in Appendix H; all experiments in this paper use the consistency reward alone.

We optimise the policy with GRPO and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all inputs in $\mathcal{X}$, with N trajectories per input; advantage normalisation and policy updates then proceed as usual. We use parameter-efficient fine-tuning (LoRA) throughout.

Figure 1 depicts the method on $\mathcal{X} = \{\text{biased}, \text{unbiased}\}$ with $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$.

3.1. Properties

Avoiding obfuscation pressure. The RMCT reward depends only on the trait T (and its derivatives p_x and p_{ref}) leaving the remaining output free to vary. Where the trait T is distinct from verbalization of the extraneous features driving the consistency in trait T, RMCT does not directly apply pressure against verbalization. Output-only RL supervision is not, however, a guarantee against obfuscation: **?** show that reasoning can still be pushed toward a safe-looking surface form via generalisation from output to chain of thought, or via token-level conditioning whereby safe-looking reasoning raises the probability of safe-looking outputs and is reinforced indirectly.

Flexibility. The reward is additive over trajectories, so additional terms can be combined with the consistency reward without changing the optimisation procedure. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight.

Output perturbations. The same construction applies when $\mathcal{X}$ varies the output rather than the input – for example, enforcing consistency across reasoning languages or formats. Whether obfuscation pressure is genuinely absent in this regime requires separate study, since output-side perturbations constrain the chain of thought directly.

Flexible reference structure. The target $p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}})$ is determined by two choices: a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ and an aggregation function Q. The method therefore does not require a privileged "clean" input. Setting $|\mathcal{X}_{\text{ref}}| = 1$ recovers a directional objective that imitates one designated input (the regime used in our experiments); setting $\mathcal{X}_{\text{ref}} = \mathcal{X}$ recovers a symmetric objective that pulls all rates toward a common target. The choice of Q further controls which references the policy is pulled toward: $Q =$ mean pulls toward the average reference rate, $Q = \min$ toward the lowest, and so on. This permits training over arbitrary sets of perturbations, including more than two and including settings where no unperturbed reference is available.

4. Experiments

4.1. Environment and bias types

We validate the method on the sycophancy benchmark of Chua et al. (2024). The benchmark consists of multiple-choice questions to which a small prompt-level perturbation has been added that suggests a particular answer. We use six single-turn bias types from the benchmark: suggested answer, distractor fact, distractor argument, post hoc, extraneous few-shot squares, and wrong few-shot. Brief descriptions are given in Appendix A; see Chua et al. (2024) for full specifications.

4.2. Models

We run experiments on two open-weight models: Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B. For Meta Llama 3.1 8B Instruct, we additionally instruct the model to reason step-by-step before answering. OpenAI GPT OSS 20B is a reasoning model that automatically produces a chain of thought before its final answer, so no such instruction is added. All training uses LoRA adapters of rank 8.

4.3. Train/evaluation split

We train on the held-in question sources LogiQA and HellaSwag (1024 questions sampled from each, drawn from the test split following the BCT data-generation procedure) and evaluate on a held-out question source, the multiple-choice subset of Humanity's Last Exam (HLE), with 100 questions per bias type per checkpoint and 400 questions per bias type for the base-model reference. For BCT, an additional 2048 instruction-following samples (prompts drawn from the Cleaned Alpaca dataset, with completions re-sampled from the target model so the supervision is self-distilled) are mixed into the training data to mitigate degradation on unrelated tasks. We train on the distractor-argument bias only and evaluate on all six bias types, which lets us measure transfer of the consistency objective to unseen biases. An additional regime that trains on a mixture of distractor-argument and wrong-few-shot prompts is reported in Appendix G.

4.4. Training settings

BCT is fine-tuned by SFT for one epoch on the 2048 BCT samples plus the 2048 instruction-following samples, at batch size 128 (32 optimisation steps). RMCT is trained with GRPO for one epoch over 64 prompts at batch size 4 prompts per update (16 update steps); each prompt is expanded into a group of $|\mathcal{X}| = 2$ inputs (biased and unbiased) with $N = 128$ rollouts per input, and the consistency reward uses the unbiased prompt as the reference. For each method-model combination we run three learning rates and pool the resulting checkpoints when reporting metrics, which gives a binomial standard error without tripling the rollout budget. Remaining hyperparameters are listed in Appendix B.

4.5. Metrics

For each evaluation question x we sample one response under the biased prompt and one under the unbiased prompt, and let $b(x), u(x) \in \{0, 1\}$ indicate whether the model selected the biased option on each. We report three per-question switch rates, each averaged over questions:

$$\text{BSR}_{\rightarrow} = \tfrac{1}{|X|} \textstyle\sum_x \max(0,\, b(x) - u(x)),$$
$$\text{BSR}_{\leftarrow} = \tfrac{1}{|X|} \textstyle\sum_x \max(0,\, u(x) - b(x)),$$
$$\text{BSR}_{\text{tot}} = \tfrac{1}{|X|} \textstyle\sum_x |b(x) - u(x)|,$$

which we call the *towards-bias switch rate*, the *away-from-bias switch rate*, and the *total switch rate* respectively. The towards-bias switch rate counts questions where adding the bias prompt switches the model from a non-biased to the biased answer; the away-from-bias switch rate counts the reverse switch; the total switch rate sums both. Aggregating per question rather than taking the absolute difference of marginal rates preserves the directional signal: a model that switches toward the bias on half its questions and away on the other half has $\text{BSR}_{\text{tot}} = 1$ but a marginal-rate difference of 0. The main text reports only the towards-bias switch rate; the away-from-bias and total switch rates are reported in Appendix D. We also report the *Bias Verbalisation Rate* (BVR), the rate at which the model's chain of thought verbalises the perturbation (scored by Google Gemma 4 31B); a drop in BVR after consistency training indicates obfuscation.

4.6. Baselines

We compare RMCT against three baselines: (i) the untrained base model; (ii) BCT trained on a matched training set (same question sources, same training bias regime); and (iii) for each of BCT and RMCT, a *control* variant trained on the same questions but with the bias-inducing perturbation removed: the model is trained for consistency between two

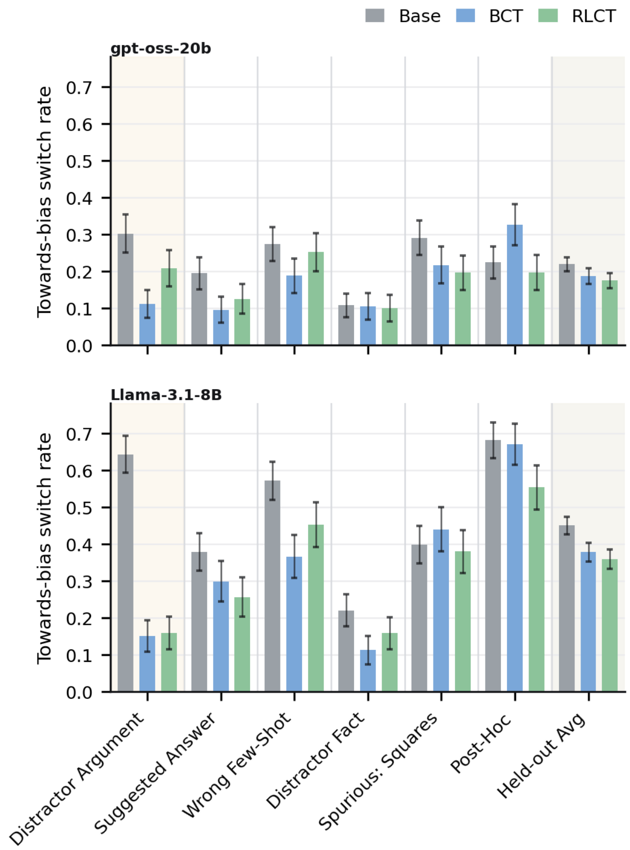

Figure 2. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

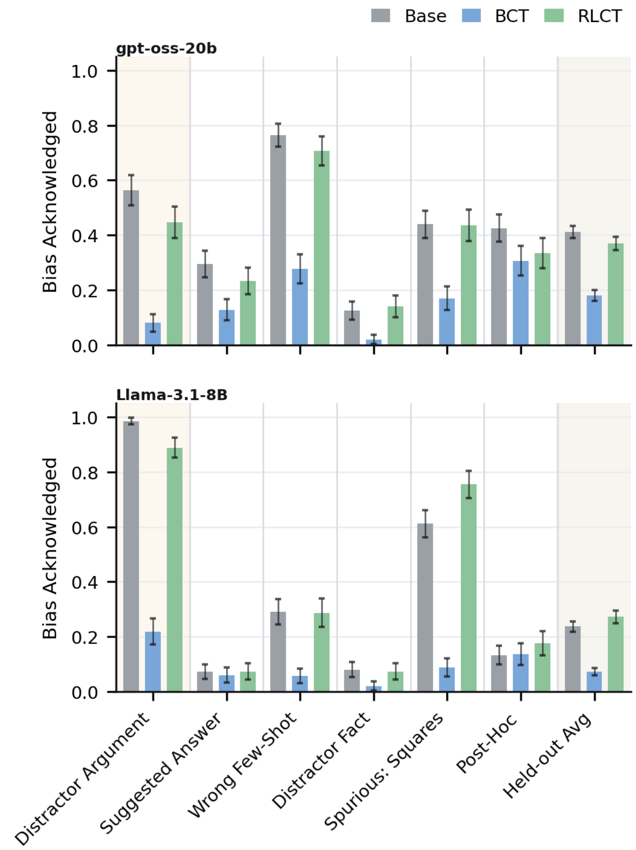

Figure 3. Bias verbalisation rate (BVR) on HLE under the biased prompt. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

unbiased copies of the prompt rather than between the biased and unbiased prompt. The control isolates training-induced drift from the consistency signal: any change in the switch rates or BVR observed in the control reflects the effect of fine-tuning on unbiased data alone, leaving the residual gap between the control and the consistency-trained run as the contribution of the consistency objective itself. Control results are reported in Appendix C.

4.7. Results

Figure 2 reports the towards-bias switch rate ($\mathrm{BSR}_{\rightarrow}$) on the held-out HLE evaluation set; Figure 3 reports the bias verbalisation rate (BVR). Results for the DA+WFS training regime, the matched controls, the away-from-bias and total switch rates, the verbalised/unverbalised splits, and accuracy are reported in the appendix.

Bias following. On the training bias both methods substantially reduce the towards-bias switch rate relative to the base model. On Meta Llama 3.1 8B Instruct, BCT reduces $\mathrm{BSR}_{\rightarrow}$ from 0.64 to 0.15 and RMCT reduces it to 0.16; on OpenAI GPT OSS 20B, BCT reduces $\mathrm{BSR}_{\rightarrow}$ from 0.30 to 0.11 and RMCT reduces it to 0.21. RMCT therefore matches BCT's in-distribution reduction on Meta Llama 3.1 8B Instruct but reduces $\mathrm{BSR}_{\rightarrow}$ less than BCT on OpenAI GPT OSS 20B. Improvements transfer to the five held-out bias types but not uniformly: BCT decreases $\mathrm{BSR}_{\rightarrow}$ more than RMCT on wrong-few-shot for Meta Llama 3.1 8B Instruct and on suggested-answer for both models, while RMCT decreases $\mathrm{BSR}_{\rightarrow}$ more than BCT on post-hoc for both models, where BCT increases the switch rate above base on OpenAI GPT OSS 20B from 0.23 to 0.33. Averaged over the five held-out biases the two methods are within standard error of each other. On Meta Llama 3.1 8B Instruct, BCT reaches 0.38 and RMCT reaches 0.36, against a base rate of 0.45; on OpenAI GPT OSS 20B, BCT reaches 0.19 and RMCT reaches 0.18, against a base rate of 0.22.

Bias verbalisation. The two methods have markedly different effects on BVR. BCT reduces BVR on the training bias from 0.99 to 0.22 on Meta Llama 3.1 8B Instruct and from 0.56 to 0.08 on OpenAI GPT OSS 20B. This reduction transfers to the held-out biases, where the average BVR drops from 0.24 to 0.07 on Meta Llama 3.1 8B Instruct and from 0.41 to 0.18 on OpenAI GPT OSS 20B. RMCT does not measurably reduce BVR. On the training bias, RMCT reaches 0.89 on Meta Llama 3.1 8B Instruct and 0.45 on OpenAI GPT OSS 20B, against base rates of 0.99 and 0.56 respectively; the held-out averages are 0.27 on Meta Llama 3.1 8B Instruct and 0.37 on OpenAI GPT OSS 20B, both within standard error of the base rates of 0.24 and 0.41. The largest between-method gap is on extraneous-squares for Meta Llama 3.1 8B Instruct, where BCT reduces BVR from 0.61 to 0.09 while RMCT moves it to 0.76.

5. Discussion

Recap. We introduced Rate Matching Consistency Training (RMCT), a reinforcement-learning method for selective consistency training that enforces consistency over a designated behavioural property without constraining other aspects of the model's output, including the chain of thought. On the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RMCT achieves bias-following reductions on held-out bias types comparable to those obtained by BCT, while preserving the rate at which the chain of thought verbalises the prompt cue close to the base-model rate. BCT, by contrast, substantially reduces this rate on both the trained and the held-out biases.

Interpretation of results. We interpret the divergent behaviour of the two methods on bias verbalisation as direct empirical evidence of the obfuscation pressure that motivates selective consistency training. Under BCT, supervising the model to reproduce a clean-prompt response on a biased input penalises any output that deviates from that target, including any verbalisation of the cue; the resulting suppression of verbalisation, of approximately 56–70% on the held-out average, is precisely the obfuscation that the method's design predicts. Under RMCT the reward depends only on the per-perturbation behaviour rates and so does not penalise the chain of thought regardless of its content; the verbalisation rate accordingly remains within standard error of the base model in both training regimes. The held-out transfer of the BVR collapse under BCT indicates that the obfuscation pressure is not specific to the trained bias type. A practical consequence is that a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RMCT.

Comparison with the original BCT results. The reductions in $\mathrm{BSR}_{\rightarrow}$ that we obtain under BCT are smaller than those reported by Chua et al. (2024). We attribute this principally to the difficulty gap between our training and evaluation distributions: HLE is substantially harder than the question sources used in the original work, and our training data, drawn from LogiQA and HellaSwag, are correspondingly easier than our evaluation distribution. Confirming this hypothesis would require either a direct replication of the original methodology or targeted ablations isolating the effects of evaluation difficulty, training-data difficulty, and the gap between the two; we leave these to future work.

In-distribution bias-following on OpenAI GPT OSS 20B. On the trained distractor-argument bias, RMCT reduces $\mathrm{BSR}_{\rightarrow}$ less than BCT on OpenAI GPT OSS 20B (0.21 versus 0.11, against a base rate of 0.30), whereas on Meta Llama 3.1 8B Instruct the two methods reach the same reduction within standard error. We offer three non-exclusive explanations for the smaller in-distribution reduction on OpenAI GPT OSS 20B. First, the comparatively low base rate of bias-following on this model yields advantages of correspondingly small magnitude under the GRPO objective, slowing optimisation. Second, the RMCT regime in our configuration trains for only sixteen update steps, which may be insufficient to reach the floor accessible on this model. Third, the initial policy may already lie close to a local optimum on this bias, leaving limited room for the consistency reward to act. Disentangling these explanations would require a scaling study over training steps together with an examination of the advantage statistics during training.

Out-of-distribution transfer. Despite the smaller in-distribution reduction on OpenAI GPT OSS 20B, the average held-out $\mathrm{BSR}_{\rightarrow}$ obtained by RMCT is within standard error of that obtained by BCT on this model. We record this as an empirical observation and do not attempt to identify its mechanism in the present work.

Post-hoc misgeneralisation under BCT. Among the held-out bias types, post-hoc is the only one expressed across multiple turns; the remaining biases are single-turn perturbations of the prompt. On OpenAI GPT OSS 20B, a reasoning model, BCT increases $\mathrm{BSR}_{\rightarrow}$ above the base rate on post-hoc (from 0.23 to 0.33), whereas RMCT does not. On Meta Llama 3.1 8B Instruct, a non-reasoning model, BCT does not exhibit this misgeneralisation, and the original BCT results of Chua et al. (2024) – obtained on non-reasoning models – likewise report no such effect. The pattern indicates that the misgeneralisation is specific to the combination of BCT and reasoning models on multi-turn cues; a detailed investigation is beyond the scope of this work.

Methodological choices. We use six of the bias types from Chua et al. (2024) and exclude three. The "Are you sure?" and positional biases are scored by a procedure that differs from the per-question switch rate used in the present work, which would preclude a uniform interpretation of $\mathrm{BSR}_{\rightarrow}$ and BVR across bias types; see Chua et al. (2024) for details. Hindsight is supported only on a single source dataset that we did not adopt, since we wished to evaluate on a more recent and harder benchmark, namely HLE. For each method-model configuration we train at three learning rates and pool checkpoints across them when computing metrics; this yields a binomial standard error on the headline metrics without tripling the rollout budget required to obtain it, and forecloses learning-rate cherry-picking. The remaining configuration choices – the BCT supervised-sample budget, the inclusion of the auxiliary instruction-following samples and the BCT batch size; and the RMCT rollout count, batch size, and KL-penalty coefficient – were selected on hyperparameter-tuning splits prior to the held-out evaluations reported here. These tuning splits do not overlap with the training or evaluation datasets used for the reported experiments, and post-hoc was excluded from the tuning evaluations so as not to inflate its apparent transfer.

Data efficiency vs. compute. RMCT extracts a learning signal from sample-level variation within each group of trajectories, without requiring a curated clean-prompt target response per training example as BCT does. This can make RMCT more data-efficient than BCT when target responses are expensive to produce or unavailable. The tradeoff is compute: each training input requires sampling $|\mathcal{X}| \cdot N$ trajectories per update, making RMCT substantially more computationally heavy per step than supervised BCT.

Limitations and future work. Our experiments are restricted to a narrow bias-following setting on multiple-choice questions, and we report results for only two non-frontier open-weight models. Future work will examine generalisation along both axes: of the method to other inconsistency problems, such as jailbreak susceptibility, persona drift, and evaluation gaming; and of the trained models to more capable, frontier-scale LLMs. The same construction also applies to consistency across input or output modalities and to other behavioural properties of the response, which we likewise leave to future work.

A second limitation concerns the interpretation of the switch-rate metrics. $\mathrm{BSR}_{\rightarrow}$, $\mathrm{BSR}_{\leftarrow}$, and $\mathrm{BSR}_{\mathrm{tot}}$ measure changes in the parsed final answer between the biased and unbiased prompts, and so conflate two distinct mechanisms: a genuine influence of the cue on the model's reasoning or internal activations, and sampling variance under which the model switches toward or away from the biased option irrespective of the cue. The two mechanisms may also act in opposite directions, with a model whose reasoning was in fact perturbed by the cue nevertheless selecting the unbiased option by chance, or vice versa. Eliminating this confound requires not only fixed sampling seeds but full batch-invariant inference (He & Thinking Machines Lab, 2025), which we leave to future work.

A third limitation concerns our definition of bias verbalisation, which counts as a verbalisation any chain-of-thought content the model could not have produced if the cue were absent from its context. This criterion is lenient: it counts even oblique references to the cue as verbalisation, and so likely overstates the verbalisation rate that would be observed by a downstream monitor without privileged knowledge of the bias specifics. Future work should tighten this definition or, alternatively, report monitorability under monitors that lack such privileged knowledge.

6. Conclusion

We introduced Rate Matching Consistency Training (RMCT), a method for selective consistency training that enforces consistency only over a designated behavioural property of the response and leaves the rest of the output, including the chain of thought, unconstrained. On the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RMCT matches BCT's reduction in towards-bias switching on held-out bias types while preserving the rate at which the chain of thought verbalises the prompt cue close to the base-model rate; BCT, by contrast, substantially suppresses verbalisation across both the trained and the held-out biases. We read this divergence as direct empirical evidence of the obfuscation pressure that motivates selective consistency training: a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RMCT. By severing the link between behavioural consistency and chain-of-thought content, RMCT offers a path to consistency training that does not trade behavioural robustness against monitorability.

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A. Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see Chua et al. (2024) for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., "My observations typically align with B."). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An "irrelevant fun fact" about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Distractor argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as "someone else's argument I read", and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., "The best answer is: (B)"); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

extraneous few-shot squares. Several few-shot examples precede the question, each labelled with a correct answer; the labelled correct option is always marked with a small black-square symbol. A model that picks up the extraneous symbol-to-label association will pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistently incorrect labelling.

B. Training hyperparameters

All training uses LoRA adapters of rank 8.

BCT. A midway checkpoint is saved every 16 steps.

RMCT. Rollouts are sampled at temperature 1.0 with up to 20,480 generated tokens. The KL penalty to a fixed reference policy is 0.05 and the policy update uses the standard PPO clipped objective.

Learning-rate pooling. For each method-model combination we run three learning rates $\{1.0,\ 2.86,\ 5.0\} \times 10^{-4}$ and pool checkpoints across them when computing metrics. This gives a binomial standard error on the metric without tripling the rollout budget.

C. Control results

The control variants – BCT trained against an unbiased target and RMCT trained with two unbiased copies of the prompt – are shown alongside the consistency-trained runs as hatched bars (**BCT Control**, **RMCT Control**) in every appendix figure. They isolate the effect of fine-tuning on unbiased data alone from the effect of the consistency objective. Across the switch-rate plots (Figures 4–10) the controls track the base model closely on most held-out biases and sit visibly above the consistency-trained runs on the trained bias, confirming that the trained-bias reduction is driven by the consistency signal rather than by drift from fine-tuning.

The controls also rule out an obfuscation null hypothesis for BCT. Controls preserve bias verbalisation near the base rate (Figure 11), so the BVR collapse seen for BCT proper is specifically attributable to training the model to match a clean-prompt response on biased inputs. The RMCT controls likewise leave BVR near base, ruling out the alternative that RMCT happens to boost verbalisation for unrelated reasons.

D. Switch-rate breakdowns

This appendix reports the towards-bias, away-from-bias, and total per-question switch rates under both the DA-only and the DA+WFS training regimes (top and bottom rows of every figure), as well as the verbalised and unverbalised splits of the towards-bias and total switch rates, broken down by which model produces which split.

D.1. Towards-bias, away-from-bias, total

Figures 4, 5, and 6 report $\mathrm{BSR}_{\rightarrow}$, $\mathrm{BSR}_{\leftarrow}$, and $\mathrm{BSR}_{\mathrm{tot}}$ for both models. The DA-only towards-bias rates (Figure 4, top rows) reproduce the main-text plot with controls overlaid; the DA+WFS rows (bottom) extend the picture to a richer training mix.

For both models the away-from-bias rate is small relative to the towards-bias rate, with $\mathrm{BSR}_{\leftarrow} \lesssim 0.15$ for almost every cell, so the total switch rate is dominated by towards-bias switching and Figure 6 largely tracks Figure 4. Two patterns are worth flagging. First, on Meta Llama 3.1 8B Instruct, BCT raises $\mathrm{BSR}_{\leftarrow}$ on the trained DA bias to roughly twice the base rate while RMCT leaves it unchanged: BCT trains a directional response, namely always pick the unbiased answer, rather than a property of consistency. Second, on OpenAI GPT OSS 20B, $\mathrm{BSR}_{\leftarrow}$ stays near base for both methods, so the small reduction in $\mathrm{BSR}_{\rightarrow}$ for RMCT relative to the base translates into a similarly small reduction in $\mathrm{BSR}_{\mathrm{tot}}$.

The DA+WFS regime (bottom rows) lowers $\mathrm{BSR}_{\rightarrow}$ further on the two trained biases for both methods, with held-out averages within standard error of the DA-only regime: BCT reduces $\mathrm{BSR}_{\rightarrow}$ on wrong-few-shot to within standard error of zero for both models, RMCT reduces wrong-few-shot $\mathrm{BSR}_{\rightarrow}$ by a smaller amount, and the held-out averages remain within standard error of each other.

D.2. Verbalised and unverbalised splits

We split each question by whether the model verbalised the bias in its chain of thought on the biased prompt. The verbalised and unverbalised subsets per condition have very different sample sizes: BCT verbalises rarely, so the verbalised plot has small n for BCT, while the unverbalised plot has small n for RMCT and the base model on biases the model habitually verbalises, for example distractor argument on Meta Llama 3.1 8B Instruct, where the base verbalises in over 99% of cases. The per-bar n counts annotated on the figures should be consulted before reading too much into any individual cell. We show the splits only for the towards-bias rate in Figures 7 and 8, and the total switch rate in Figures 9 and 10; the away-from-bias splits show the same qualitative picture and are omitted to save space.

Verbalised subset. Figure 7 restricts to questions where the biased-prompt chain of thought verbalised the cue. For RMCT and the base model the verbalised subset still shows substantial bias following. On Meta Llama 3.1 8B Instruct the held-out-average towards-bias rate is 0.49 for RMCT and 0.45 for the base; on OpenAI GPT OSS 20B the rate is 0.21 for both. These rates are indistinguishable from the unconditional rates in the main text, ruling out the alternative that the RMCT reduction in $\mathrm{BSR}_{\rightarrow}$ comes from the model noticing the bias more often and only switching when it did not: even when it does notice, it follows the bias at the base rate and is moved by the reward toward consistency. For BCT the verbalised subset is too small to read off reliable per-bias rates, with $n \lesssim 30$ for several biases. Figure 9, the

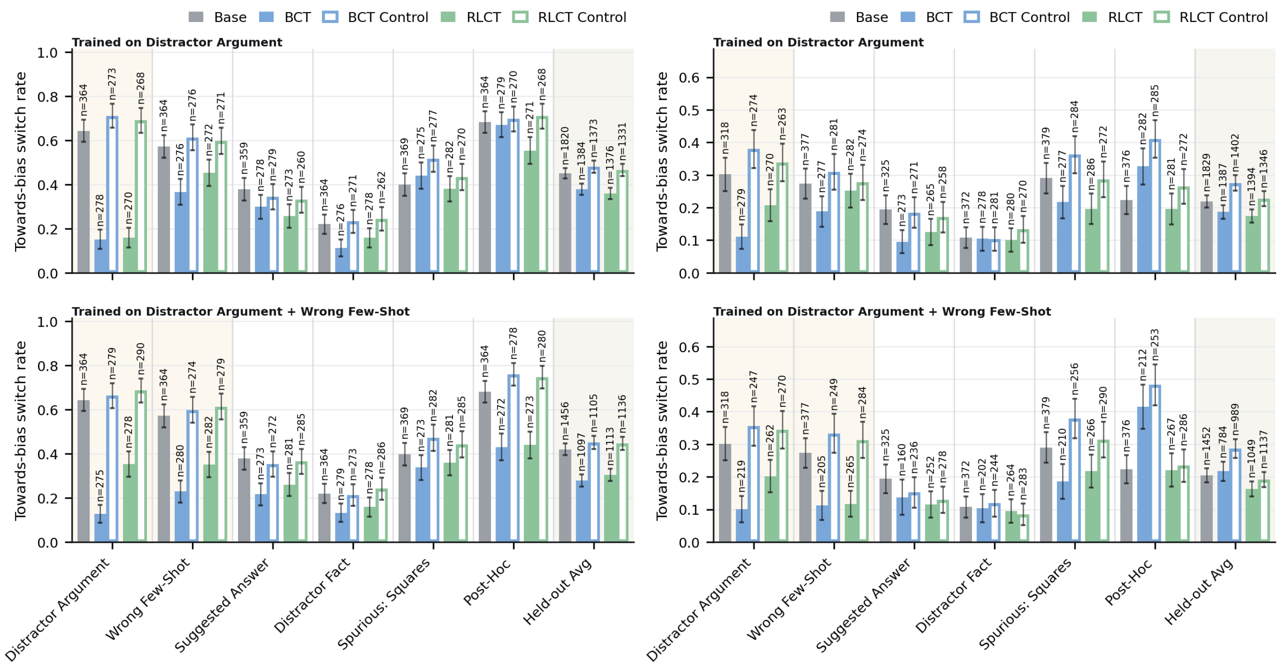

Figure 4. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

total switch rate on the same subset, shows the same pattern.

Unverbalised subset. Figure 8 restricts to questions where the chain of thought did not verbalise the cue. This is the operating regime for BCT across most biases, and indeed the BCT bars in this plot are similar to its bars in Figure 4. On biases where the base model unverbalises rarely, namely distractor argument and extraneous-squares on Meta Llama 3.1 8B Instruct, the unverbalised subset for the base model has very low n but a high $\mathrm{BSR}_{\rightarrow}$, so the few times the model fails to verbalise the cue it is much more likely to follow it. RMCT in this subset behaves similarly to BCT: when the model has not verbalised the cue, both methods reduce following at roughly comparable rates. The total switch rate (Figure 10) again mirrors the towards-bias plot.

The combination of these two splits supports our reading of the main-text BVR result. RMCT preserves verbalisation and reduces following both when the cue is verbalised and when it is not; BCT reduces following primarily by routing the model into the unverbalised regime.

E. Bias verbalisation details

Figure 11 reports the unconditional bias verbalisation rate (BVR) under the biased prompt for both models and both training regimes, and Figure 12 restricts BVR to the *bias-switched* subset: questions on which adding the bias prompt changed the parsed final answer (i.e. where $b(x) \neq u(x)$). This is the conditional BVR on questions where the bias actually moved the model.

Figure 11 reproduces the main-text result at higher resolution and across both training regimes: BCT reduces BVR on the trained bias and on the held-out average, while RMCT does not move BVR away from the base rate by more than a binomial standard error in either training regime. The DA+WFS BCT run yields a larger BVR reduction on its trained biases than the DA-only run; DA+WFS RMCT remains within standard error of base.

Figure 12 is the more behaviourally meaningful view: it asks whether, on the questions where adding the bias did swing the model's answer, the model said so in its reasoning. On Meta Llama 3.1 8B Instruct, the held-out-average conditional BVR is 0.30 for the base model and 0.06 under BCT, but stays at 0.30 under RMCT; on OpenAI GPT OSS 20B it drops from 0.42 to 0.20 under BCT and stays at 0.41 under RMCT. Equivalently, under BCT, on the questions where the bias was operative, the model now reasons as if no bias were present in approximately 80% of cases on Meta Llama 3.1 8B Instruct and approximately 50% of cases on OpenAI GPT OSS 20B, whereas under RMCT this rate is within standard error of the base rate. An verbalisation-rate

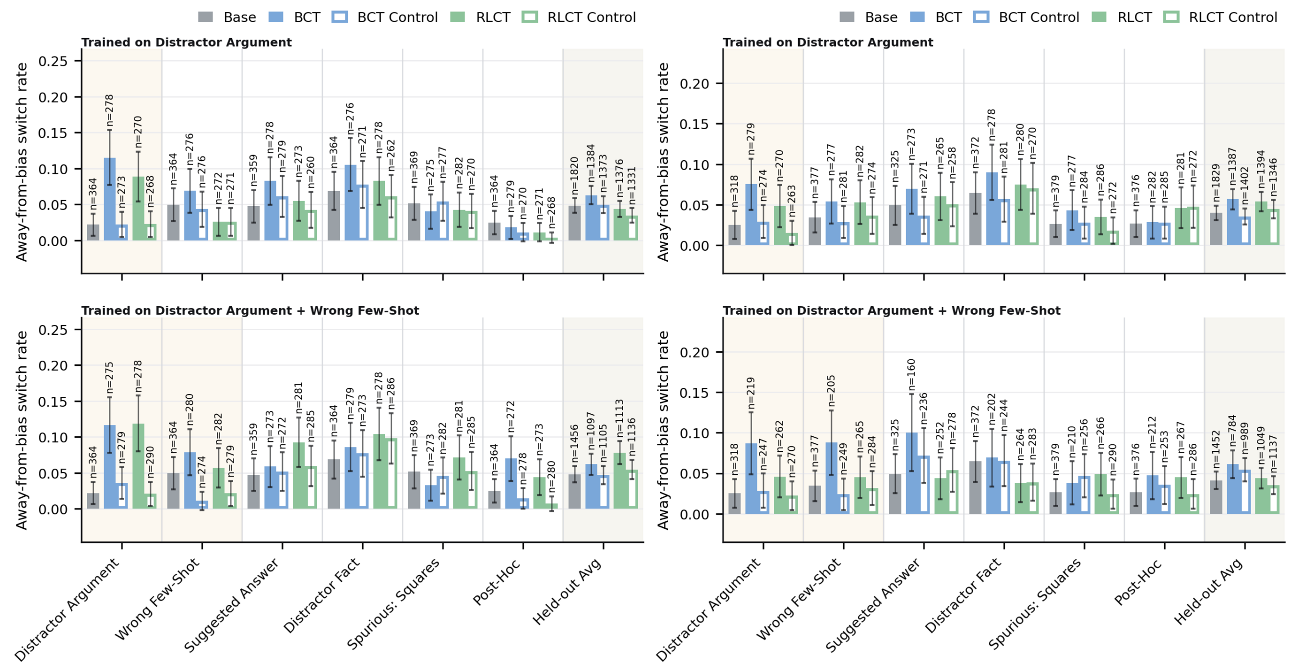

Figure 5. Away-from-bias switch rate $\mathrm{BSR}_{\leftarrow}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

monitor calibrated on the base model would therefore fail to flag the bias-driven answers under BCT, but not under RMCT.

F. Accuracy

We report two accuracy metrics: accuracy under the biased prompt in Figure 13, and accuracy under the unbiased prompt in Figure 14. HLE is intentionally hard, so absolute accuracies remain at $\lesssim 0.25$ throughout.

Under the biased prompt, BCT and RMCT both leave accuracy within standard error of the base model on Meta Llama 3.1 8B Instruct and on OpenAI GPT OSS 20B across all held-out biases and both training regimes. Where the consistency-trained runs reduce $\mathrm{BSR}_{\rightarrow}$, they do not do so by trading correctness in either direction. Under the unbiased prompt the differences are small but consistent. On Meta Llama 3.1 8B Instruct, RMCT reaches 0.18, BCT reaches 0.16, and the base reaches 0.13; on OpenAI GPT OSS 20B, BCT reaches 0.14, RMCT reaches 0.10, and the base reaches 0.11. In every case the differences sit within the binomial standard error. The unbiased-prompt bias-match rate in Figure 15 is reported for completeness: on the unbiased prompt, RMCT does not raise the rate at which the model picks the bias-target option above base, so the bias-following reduction is not produced by inducing a generic preference against those options.

G. Distractor-argument + wrong-few-shot training regime

We additionally trained BCT and RMCT on a mixture of distractor-argument and wrong-few-shot prompts (DA+WFS) on LogiQA+HellaSwag, with all other settings matching the DA-only setup of Section 4. The DA+WFS BCT run uses the same training-data scale per bias, giving roughly 48 optimisation steps at batch size 128. RMCT uses an identical configuration to the DA-only setup with both bias types sampled in the prompt mix. Per-metric DA+WFS results are shown as the second row of every figure in Appendices D, E, and F. The qualitative picture matches the DA-only regime: BCT collapses BVR on its trained biases while RMCT preserves it, and the held-out switch-rate reductions are comparable between the two methods.

H. Anchor reward

To prevent the target rate p_{ref} from drifting as the policy updates, RMCT supports an optional anchor reward that ties each reference rate p_x ($x \in \mathcal{X}_{\mathrm{ref}}$) to its initial value $p_x^{(0)}$ computed under the initial policy π_{θ_0} (a checkpoint taken

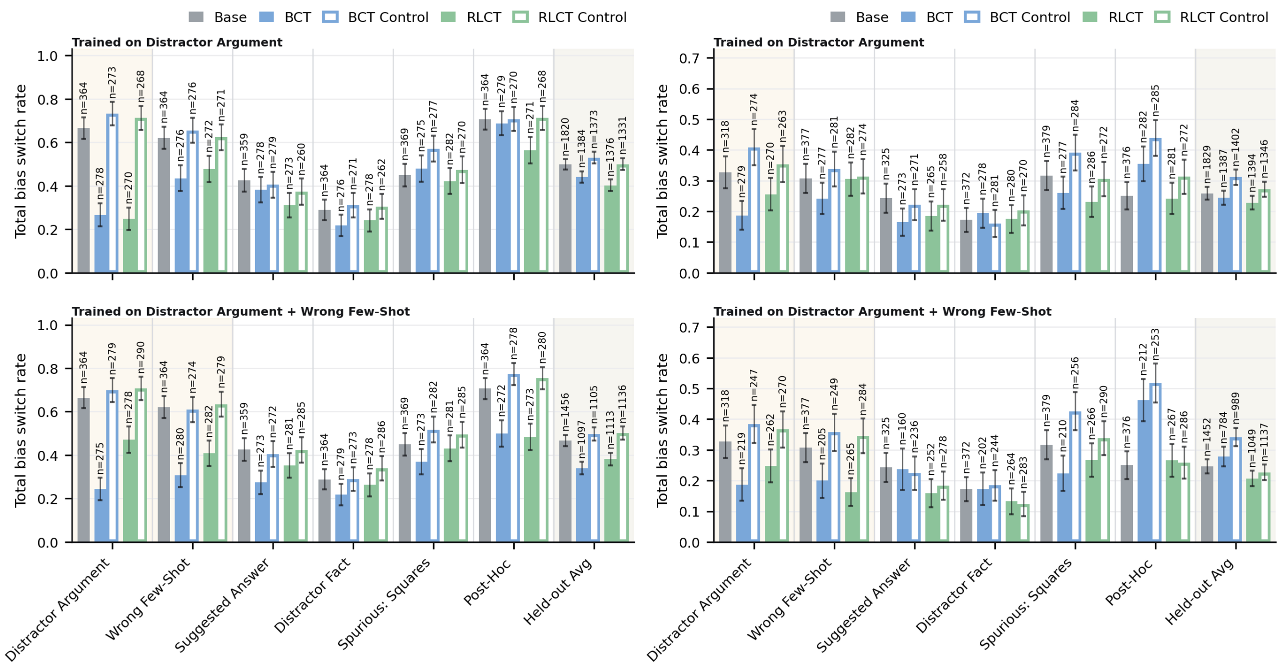

Figure 6. Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

before consistency training begins):

$$r^{\mathrm{anchor}}_{x,i} = -(p_x - p^{(0)}_x)\,(T(\tau_{x,i}) - p_x) \qquad x \in \mathcal{X}_{\mathrm{ref}} \quad (6)$$

The baseline p_x is the variance-optimal centring under the current sampling distribution; $p^{(0)}_x$ comes from a frozen policy and would not yield an unbiased baseline. Anchoring each p_x individually keeps $p^{(0)}_{\mathrm{ref}} = Q(\{p^{(0)}_x\}_{x\in\mathcal{X}_{\mathrm{ref}}})$ stable under any choice of aggregator Q.

Reference and non-reference trajectories receive different reward terms, combined by a mixing weight α:

$$r_{x,i} = \begin{cases} (1-\alpha)\, r^{\mathrm{cons}}_{x,i} & x \in \mathcal{X} \setminus \mathcal{X}_{\mathrm{ref}} \\ \alpha\, r^{\mathrm{anchor}}_{x,i} & x \in \mathcal{X}_{\mathrm{ref}} \end{cases} \quad (7)$$

All experiments in this paper use $\alpha = 0$ (the consistency reward alone). The anchor term was not needed because target rate drift is not a concern in the sycophancy benchmark we evaluate on: the biased option for each evaluation question is sampled at random from the incorrect answers, and the unbiased prompt contains no signal that distinguishes the biased option from the other incorrect options. The target rate p_{ref} can therefore only drift via an overall shift in the unbiased-prompt answer distribution, for instance through a change in accuracy. Figure 14 shows a small, non-significant increase in unbiased-prompt accuracy under RMCT, which would correspond to a slight downward drift in p_{ref} – a direction that makes the consistency target more aggressive rather than degenerate, and so does not threaten the objective.

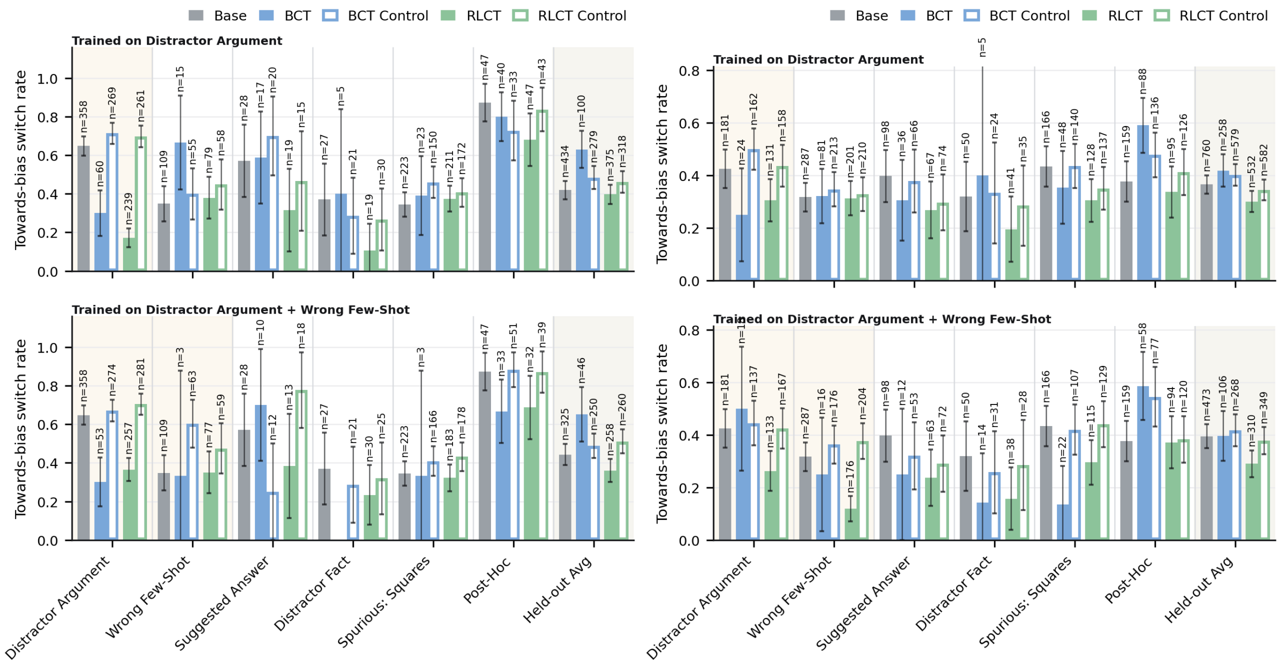

Figure 7. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

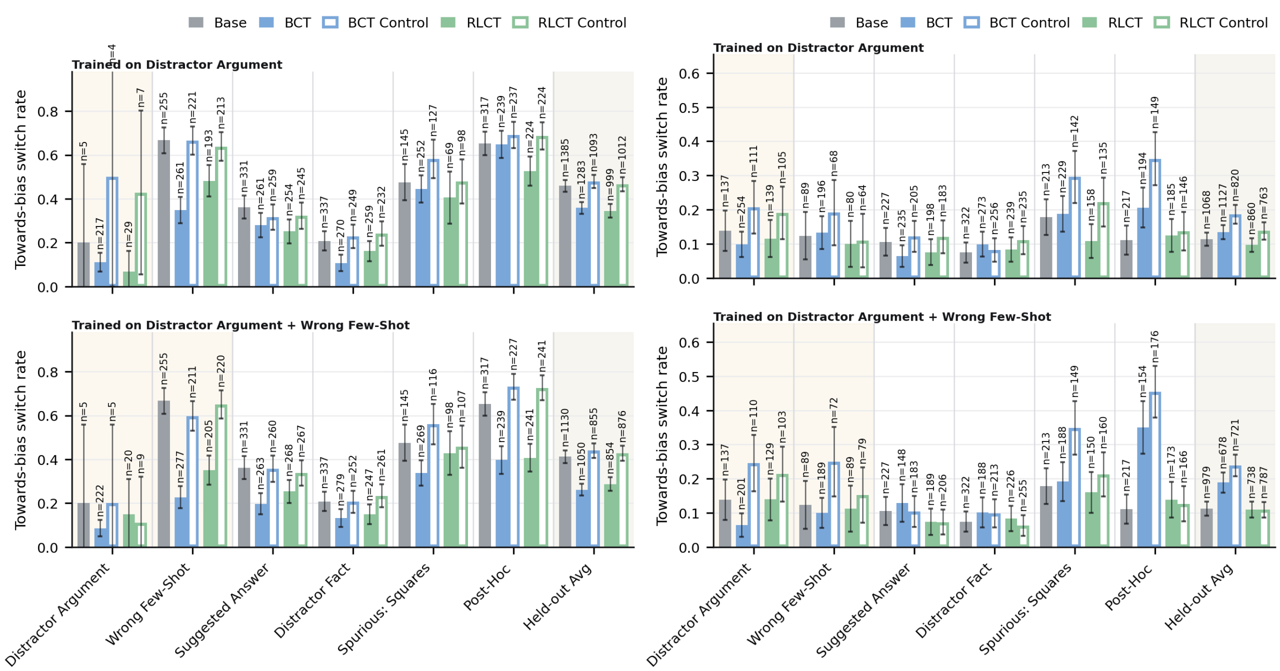

Figure 8. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

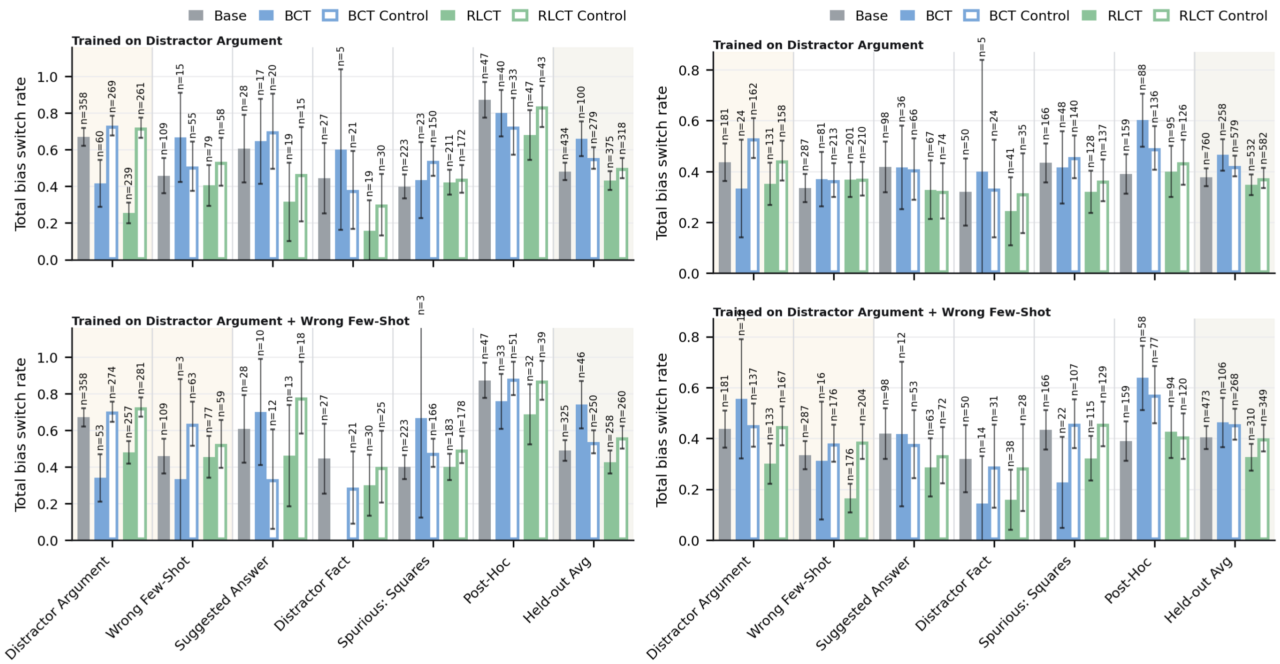

Figure 9. Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

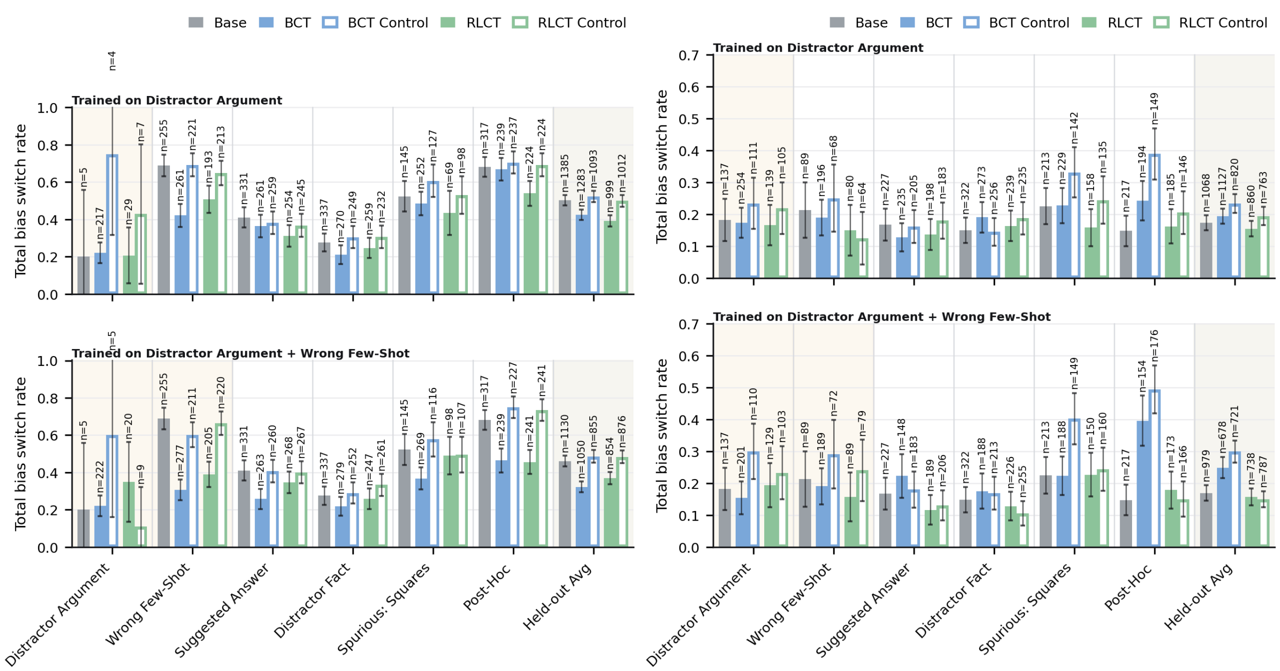

Figure 10. Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

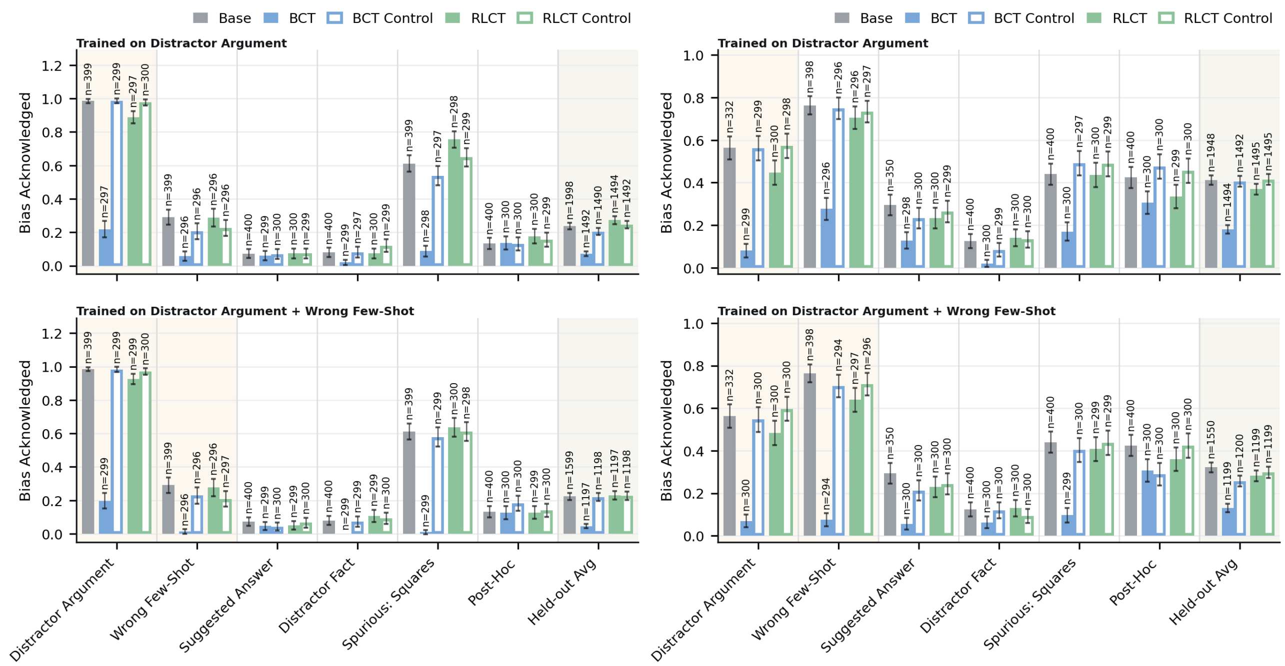

Figure 11. Bias verbalisation rate (BVR) on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

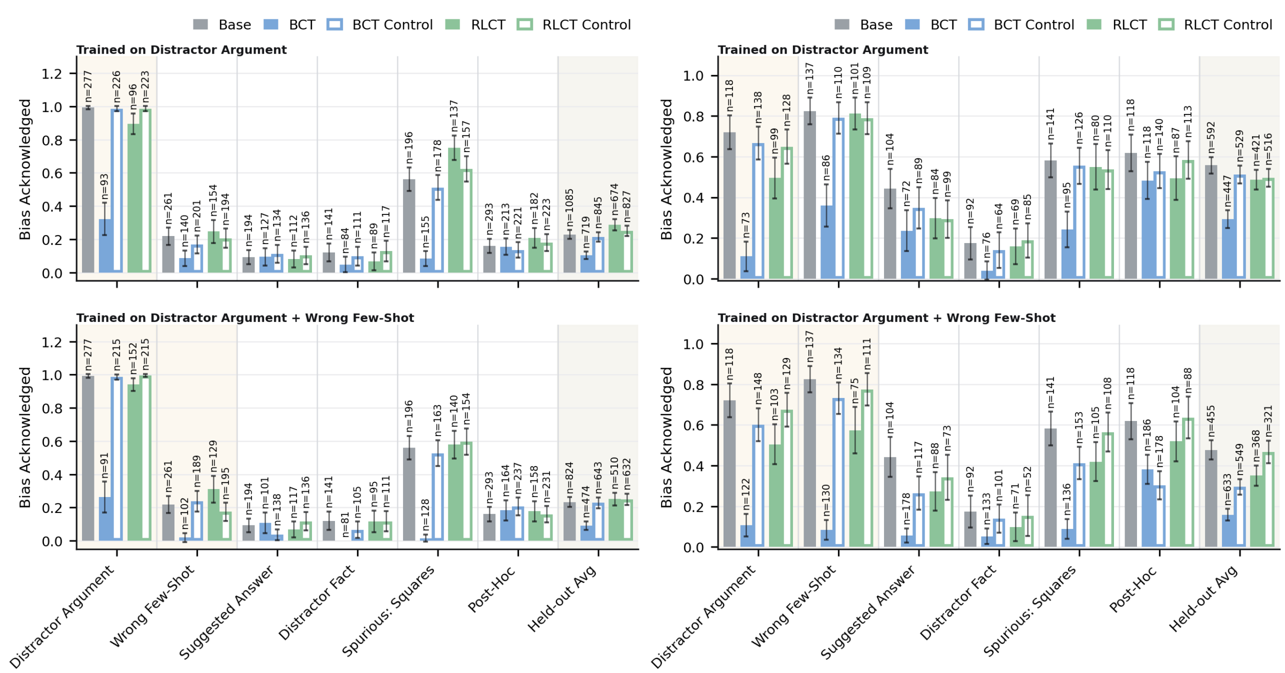

Figure 12. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions where the final answer changed between the unbiased and biased prompts. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

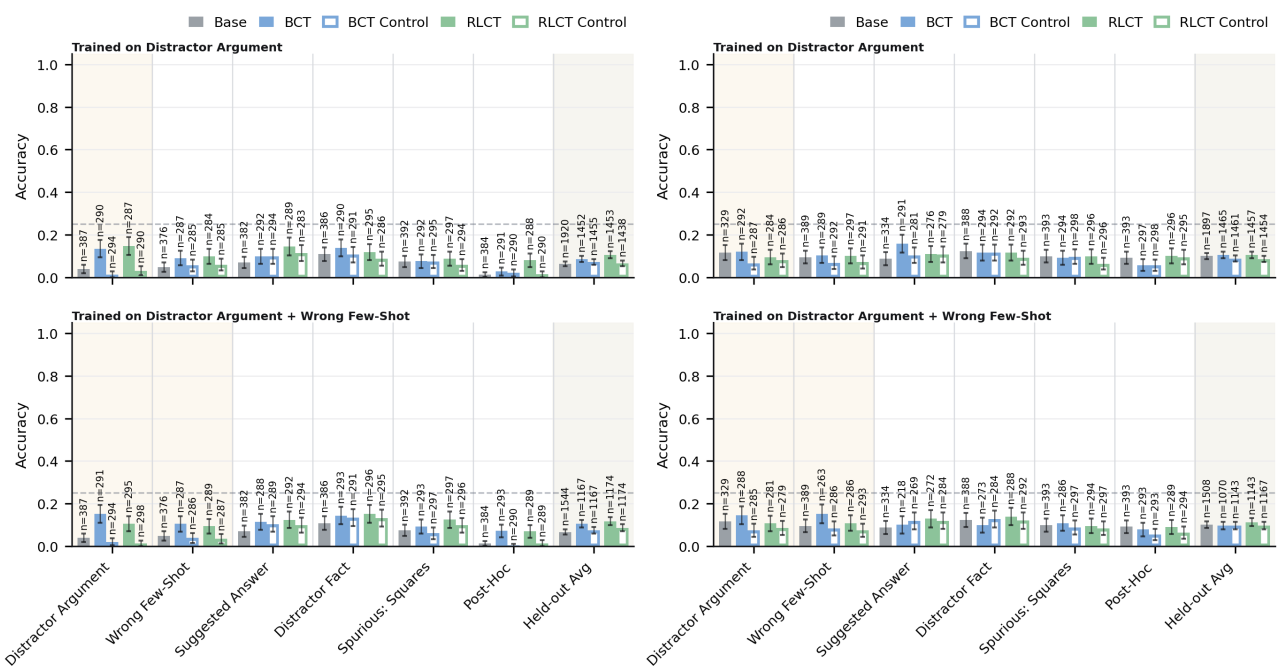

Figure 13. Accuracy on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

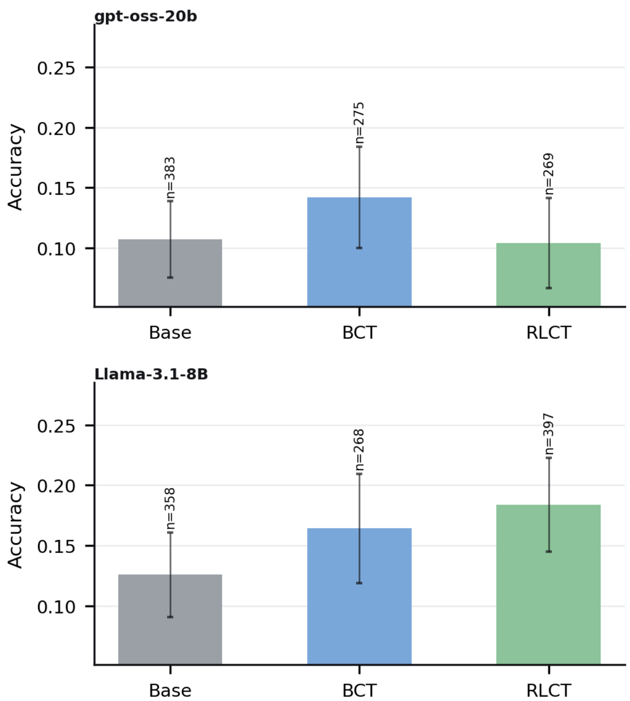

Figure 14. Accuracy on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

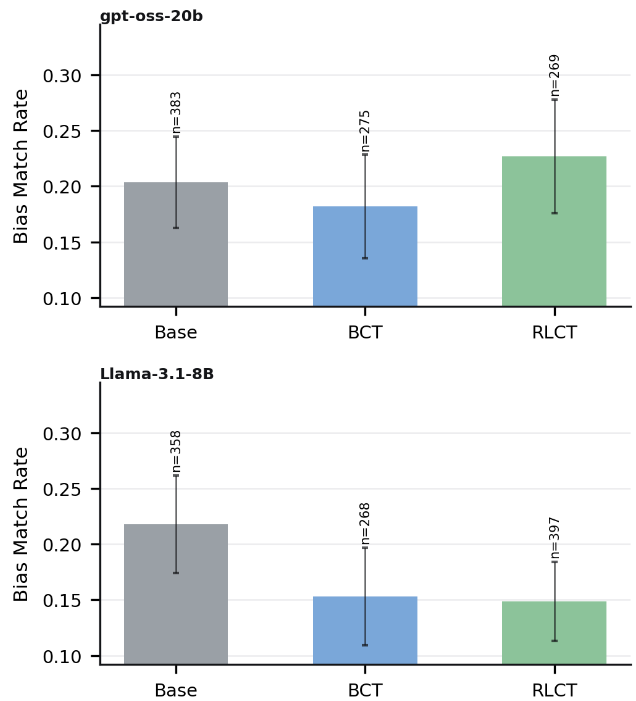

Figure 15. Bias-target match rate on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.